$$f(x) = \ln\left(\frac{2x-1}{x-1}\right)$$

$$f'(x) = -\frac{1}{(2x-1)(x-1)}$$

$f'(x) < 0$ in $\left]-\infty, \frac{1}{2}\right[\cup \left]1, +\infty\right[$

Dominio di f

Allora f è STRETTAMENTE DECRESCENTE

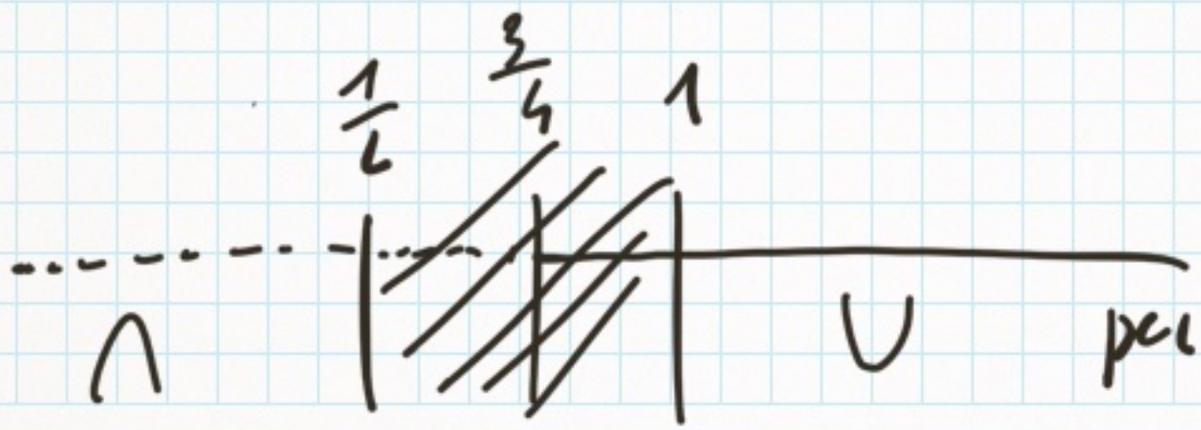
$$f'(x) = -\frac{1}{(2x-1)(x-1)} = -\frac{1}{2x^2-3x+1}$$

calcolo

$$f''(x) = -\frac{0\cdot(2x^2-3x+1) - 1\,(4x-3)}{(2x^2-3x+1)^2} =$$

$$= +\frac{4x-3}{(2x^2-3x+1)^2} > 0$$

$$f''(x) \geq 0 \iff 4x-3 \geq 0 \iff x \geq \frac{3}{4}$$



per $\frac{3}{4}$

NON CI SONO PUNTI DI FLESSO NEL DOMINIO

Integrazione per parti

Siano f, g derivabili con derivate continue,

$f, g : I \to \mathbb{R}$.

Allora

$$\int f'(x) g(x)\, dx = f(x) g(x) - \int f(x) \cdot g'(x)\, dx$$

Inoltre

$$\int_a^b f'(x) g(x)\, dx \quad [f(x) g(x)]_a^b - \int_a^b f(x) \cdot g'(x)\, dx$$

.

Integrali per ITERAZIONE

$$\int e^x \cdot \cos x \, dx$$

$f'(x) = e^x \quad f(x) = e^x$

Integrazione per parti

$g(x) = \cos x, \quad g'(x) = -\sin x$

$$\int e^x \cos x \, dx = e^x \cdot \cos x - \int e^x (-\sin x) \, dx =$$

$$= e^x \cos x + \int e^x \sin x \, dx$$

INTEGRIAMO DI NUOVO PER PARTI MANTENENDO $f'(x) = e^x$

$$\int_x^x \cos x\,dx + \int_x^x \cos x\,dx = \sin x + b_1 x \;\Rightarrow\; 2\int_x^x \cos x\,dx = \sin x + b_1 x$$

$$\int_x^x \cos x\,dx = \frac{\sin x + b_1 x}{2}$$

$$\int_x^x \cos^2 x\,dx = \int_x^x \cos x \cdot \cos x\,dx = \sin x \cos x + \int_x^x \sin^2 x\,dx$$

$$= \sin x \cos x + \int_x^x (1 - \cos^2 x)\,dx = \sin x \cos x + x - \int_x^x \cos^2 x\,dx$$

$$\int_x^x \cos^2 x\,dx = \frac{\sin x \cos x + x}{2}$$

$$\int_x^x \sin^2 x\,dx = \int_x^x (1 - \cos^2 x)\,dx = x - \frac{\sin x \cos x + x}{2} = \frac{x - \sin x \cos x}{2}$$

$$\int_x^x \sin x \cos x\,dx = \int_x^x \sin x\,d(\sin x) = \frac{\sin^2 x}{2}$$

$$\int_x^x \sin^3 x\,dx = \int_x^x \sin^2 x \cdot \sin x\,dx = \int_x^x (1 - \cos^2 x)\sin x\,dx$$

$$= -\cos x + \frac{\cos^3 x}{3}$$

$$\int_x^x \cos^3 x\,dx = \int_x^x \cos^2 x \cdot \cos x\,dx = \int_x^x (1 - \sin^2 x)\cos x\,dx$$

$$= \sin x - \frac{\sin^3 x}{3}$$

$$\int_x^x \sin x \cos^2 x\,dx = -\int_x^x \cos^2 x\,d(\cos x) = -\frac{\cos^3 x}{3}$$

$$\int_x^x \sin^2 x \cos x\,dx = \int_x^x \sin^2 x\,d(\sin x) = \frac{\sin^3 x}{3}$$

$$\int_x^x \sin^2 x \cos^2 x\,dx = \int_x^x \frac{1 - \cos 2x}{2} \cdot \frac{1 + \cos 2x}{2}\,dx = \frac{1}{4}\int_x^x (1 - \cos^2 2x)\,dx$$

$$= \frac{1}{4}\int_x^x \sin^2 2x\,dx = \frac{1}{4}\int_x^x \frac{1 - \cos 4x}{2}\,dx = \frac{x}{8} - \frac{\sin 4x}{32}$$

$$\int_x^x \sin^4 x\,dx = \int_x^x \left(\frac{1 - \cos 2x}{2}\right)^2 dx = \frac{1}{4}\int_x^x (1 - 2\cos 2x + \cos^2 2x)\,dx$$

$$= \frac{1}{4}\int_x^x \left(1 - 2\cos 2x + \frac{1 + \cos 4x}{2}\right)dx = \frac{1}{4}\int_x^x \left(\frac{3}{2} - 2\cos 2x + \frac{\cos 4x}{2}\right)dx$$

$$= \frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32}$$

$$\int_x^x \cos^4 x\,dx = \int_x^x \left(\frac{1 + \cos 2x}{2}\right)^2 dx = \frac{1}{4}\int_x^x (1 + 2\cos 2x + \cos^2 2x)\,dx$$

$$= \frac{1}{4}\int_x^x \left(1 + 2\cos 2x + \frac{1 + \cos 4x}{2}\right)dx = \frac{1}{4}\int_x^x \left(\frac{3}{2} + 2\cos 2x + \frac{\cos 4x}{2}\right)dx$$

$$= \frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32}$$

$$\int_x^x \sin^2 x \cos^2 x\,dx = \frac{x}{8} - \frac{\sin 4x}{32}$$

$$\int_x^x \sin^3 x \cos^2 x\,dx = -\int_x^x (1 - \cos^2 x)\cos^2 x\,d(\cos x) = \frac{\cos^3 x}{3} - \frac{\cos^5 x}{5}$$

$$\int_x^x \sin^2 x \cos^3 x\,dx = \int_x^x \sin^2 x (1 - \sin^2 x)\cos x\,dx = \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5}$$

$$\int_x^x \sin^3 x \cos^3 x\,dx = \int_x^x \sin^3 x (1 - \sin^2 x)\cos x\,dx = \frac{\sin^4 x}{4} - \frac{\sin^6 x}{6}$$

$$\int_x^x \sin^4 x \cos^2 x\,dx = \int_x^x \left(\frac{1 - \cos 2x}{2}\right)\left(\frac{1 - \cos 2x}{2}\right)\left(\frac{1 + \cos 2x}{2}\right)dx$$

$$= \frac{1}{8}\int_x^x (1 - \cos 2x - \cos^2 2x + \cos^3 2x)\,dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} - \cos 2x + \frac{\cos 4x}{2} + (1 - \sin^2 2x)\cos 2x\right)dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} - \cos 2x + \frac{\cos 4x}{2} + \cos 2x - \sin^2 2x \cos 2x\right)dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} + \frac{\cos 4x}{2} - \sin^2 2x \cos 2x\right)dx$$

$$= \frac{1}{8}\left(\frac{x}{2} + \frac{\sin 4x}{8} - \frac{\sin^3 2x}{6}\right)$$

$$\int_x^x \sin^2 x \cos^4 x\,dx = \int_x^x \left(\frac{1 - \cos 2x}{2}\right)\left(\frac{1 + \cos 2x}{2}\right)\left(\frac{1 + \cos 2x}{2}\right)^2 dx$$

$$= \frac{1}{8}\int_x^x (1 - \cos^2 2x)(1 + \cos 2x)\,dx$$

$$= \frac{1}{8}\int_x^x (1 + \cos 2x - \cos^2 2x - \cos^3 2x)\,dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} + \cos 2x - \frac{\cos 4x}{2} - (1 - \sin^2 2x)\cos 2x\right)dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} + \cos 2x - \frac{\cos 4x}{2} - \cos 2x + \sin^2 2x \cos 2x\right)dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} - \frac{\cos 4x}{2} + \sin^2 2x \cos 2x\right)dx$$

$$= \frac{1}{8}\left(\frac{x}{2} - \frac{\sin 4x}{8} + \frac{\sin^3 2x}{6}\right)$$

$$\int_x^x \sin^3 x \cos^4 x\,dx = -\int_x^x (1 - \cos^2 x)\cos^4 x\,d(\cos x) = \frac{\cos^5 x}{5} - \frac{\cos^7 x}{7}$$

$$\int_x^x \sin^4 x \cos^3 x\,dx = \int_x^x \sin^4 x (1 - \sin^2 x)\cos x\,dx = \frac{\sin^5 x}{5} - \frac{\sin^7 x}{7}$$

$$\int_x^x \sin^5 x \cos^2 x\,dx = \int_x^x \left(\frac{1 - \cos 2x}{2}\right)^2 \left(\frac{1 - \cos 2x}{2}\right)\sin x\,dx$$

$$= \frac{1}{8}\int_x^x (1 - 2\cos 2x + \cos^2 2x)(1 - \cos 2x)\sin x\,dx$$

$$= \frac{1}{8}\int_x^x (1 - 3\cos 2x + 3\cos^2 2x - \cos^3 2x)\sin x\,dx$$

$$= \frac{1}{8}\int_x^x \left(1 - 3\cos 2x + \frac{3(1 + \cos 4x)}{2} - (1 - \sin^2 2x)\cos 2x\right)\sin x\,dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{5}{2} - 4\cos 2x + \frac{3\cos 4x}{2} + \sin^2 2x \cos 2x\right)\sin x\,dx$$

$$= \frac{1}{8}\left(\frac{5}{2}\cos x - \frac{4}{3}\cos 3x + \frac{3}{10}\cos 5x + \frac{1}{3}\sin^3 2x \cos x\right)$$

$$\int_x^x \sin^2 x \cos^5 x\,dx = \int_x^x \left(\frac{1 - \cos 2x}{2}\right)\left(\frac{1 + \cos 2x}{2}\right)^2 \left(\frac{1 + \cos 2x}{2}\right)dx$$

$$= \frac{1}{8}\int_x^x (1 - \cos^2 2x)(1 + \cos 2x)^2\,dx$$

$$= \frac{1}{8}\int_x^x (1 + 2\cos 2x - \cos^2 2x - 2\cos^3 2x)\,dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} + 2\cos 2x - \frac{\cos 4x}{2} - 2(1 - \sin^2 2x)\cos 2x\right)dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} + 2\cos 2x - \frac{\cos 4x}{2} - 2\cos 2x + 2\sin^2 2x \cos 2x\right)dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} - \frac{\cos 4x}{2} + 2\sin^2 2x \cos 2x\right)dx$$

$$= \frac{1}{8}\left(\frac{x}{2} - \frac{\sin 4x}{8} + \frac{2\sin^3 2x}{6}\right)$$

$$\int_x^x \sin^3 x \cos^5 x\,dx = -\int_x^x (1 - \cos^2 x)\cos^5 x\,d(\cos x) = \frac{\cos^6 x}{6} - \frac{\cos^8 x}{8}$$

$$\int_x^x \sin^5 x \cos^3 x\,dx = \int_x^x \sin^5 x (1 - \sin^2 x)\cos x\,dx = \frac{\sin^6 x}{6} - \frac{\sin^8 x}{8}$$

$$\int_x^x \sin^4 x \cos^4 x\,dx = \int_x^x \left(\frac{1 - \cos 2x}{2}\right)^2 \left(\frac{1 + \cos 2x}{2}\right)^2 dx$$

$$= \frac{1}{16}\int_x^x (1 - \cos^2 2x)^2\,dx$$

$$= \frac{1}{16}\int_x^x \sin^4 2x\,dx$$

$$= \frac{1}{16}\int_x^x \left(\frac{1 - \cos 4x}{2}\right)^2 dx$$

$$= \frac{1}{64}\int_x^x (1 - 2\cos 4x + \cos^2 4x)\,dx$$

$$= \frac{1}{64}\int_x^x \left(1 - 2\cos 4x + \frac{1 + \cos 8x}{2}\right)dx$$

$$= \frac{1}{64}\int_x^x \left(\frac{3}{2} - 2\cos 4x + \frac{\cos 8x}{2}\right)dx$$

$$= \frac{3x}{128} - \frac{\sin 4x}{128} + \frac{\sin 8x}{1024}$$

$$\int_x^x \sin^2 x \cos^6 x\,dx = \int_x^x \left(\frac{1 - \cos 2x}{2}\right)\left(\frac{1 + \cos 2x}{2}\right)^3 dx$$

$$= \frac{1}{8}\int_x^x (1 - \cos^2 2x)(1 + \cos 2x)^2\,dx$$

$$= \frac{1}{8}\int_x^x (1 + 2\cos 2x - \cos^2 2x - 2\cos^3 2x)\,dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} + 2\cos 2x - \frac{\cos 4x}{2} - 2(1 - \sin^2 2x)\cos 2x\right)dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} + 2\cos 2x - \frac{\cos 4x}{2} - 2\cos 2x + 2\sin^2 2x \cos 2x\right)dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} - \frac{\cos 4x}{2} + 2\sin^2 2x \cos 2x\right)dx$$

$$= \frac{1}{8}\left(\frac{x}{2} - \frac{\sin 4x}{8} + \frac{2\sin^3 2x}{6}\right)$$

$$\int_x^x \sin^3 x \cos^5 x\,dx = -\int_x^x (1 - \cos^2 x)\cos^5 x\,d(\cos x) = \frac{\cos^6 x}{6} - \frac{\cos^8 x}{8}$$

$$\int_x^x \sin^5 x \cos^3 x\,dx = \int_x^x \sin^5 x (1 - \sin^2 x)\cos x\,dx = \frac{\sin^6 x}{6} - \frac{\sin^8 x}{8}$$

$$\int_x^x \sin^4 x \cos^4 x\,dx = \int_x^x \left(\frac{1 - \cos 2x}{2}\right)^2 \left(\frac{1 + \cos 2x}{2}\right)^2 dx$$

$$= \frac{1}{16}\int_x^x (1 - \cos^2 2x)^2\,dx$$

$$= \frac{1}{16}\int_x^x \sin^4 2x\,dx$$

$$= \frac{1}{16}\int_x^x \left(\frac{1 - \cos 4x}{2}\right)^2 dx$$

$$= \frac{1}{64}\int_x^x (1 - 2\cos 4x + \cos^2 4x)\,dx$$

$$= \frac{1}{64}\int_x^x \left(1 - 2\cos 4x + \frac{1 + \cos 8x}{2}\right)dx$$

$$= \frac{1}{64}\int_x^x \left(\frac{3}{2} - 2\cos 4x + \frac{\cos 8x}{2}\right)dx$$

$$= \frac{3x}{128} - \frac{\sin 4x}{128} + \frac{\sin 8x}{1024}$$

$$\int_x^x \sin^2 x \cos^6 x\,dx = \int_x^x \left(\frac{1 - \cos 2x}{2}\right)\left(\frac{1 + \cos 2x}{2}\right)^3 dx$$

$$= \frac{1}{8}\int_x^x (1 - \cos^2 2x)(1 + \cos 2x)^2\,dx$$

$$= \frac{1}{8}\int_x^x (1 + 2\cos 2x - \cos^2 2x - 2\cos^3 2x)\,dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} + 2\cos 2x - \frac{\cos 4x}{2} - 2(1 - \sin^2 2x)\cos 2x\right)dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} + 2\cos 2x - \frac{\cos 4x}{2} - 2\cos 2x + 2\sin^2 2x \cos 2x\right)dx$$

$$= \frac{1}{8}\int_x^x \left(\frac{1}{2} - \frac{\cos 4x}{2} + 2\sin^2 2x \cos 2x\right)dx$$

$$= \frac{1}{8}\left(\frac{x}{2} - \frac{\sin 4x}{8} + \frac{2\sin^3 2x}{6}\right)$$

$$\int e^x \cos x \, dx = \frac{1}{2}\left(e^x \cos x + e^x \sin x\right) + c$$

A CASA

$$\int e^x \sin x \, dx$$

↓ PER PARTI

$$\int \cos^2 x \, dx = \int \cos x \cdot \cos x \, dx =$$

$$f'(x) = \cos x \quad f(x) = \sin x$$

$$g(x) = \cos x \quad g'(x) = -\sin x$$

$$\int \sin x \cos x \, dx = \int \sin x \, d(\sin x) = \frac{\sin^2 x}{2} + C$$

$$\int \sin x \cos x \, dx = -\int \cos x \, d(\cos x) = -\frac{\cos^2 x}{2} + C$$

$$\int \sin x \cos x \, dx = \int (1-\cos^2 x) \cos x \, dx = \int \cos x \, dx - \int \cos^3 x \, dx$$

$$\int \sinh^2 x \, dx$$

Answer

$$\int \cosh^2 x \, dx = \frac{1}{2}(\cosh x \sinh x + x) + c$$

Prove

$$\int \cos^2 x \, dx = \cos x \sinh x + x$$

INTEGRALE DI FUNZIONI RAZIONALI

- $\int \frac{1}{2x+3}\,dx = \frac{1}{2}\int \frac{2}{2x+3}\,dx =$

$= \frac{1}{2}\int \frac{(2x+3)'}{2x+3}\,dx = \frac{1}{2}\ln|2x+3| + c$

- $\int \frac{x+1}{2x+3}\,dx$

Per prima cosa bisogna effettuare la DIVISIONE DI POLINOMI

SE IL GRADO DEL NUMERATORE È MAGGIORE O UGUALE DI QUELLO DEL DENOMINATORE

$$\frac{P(x)}{P_1(x)} = Q(x) + \frac{R(x)}{P_1(x)}$$

grado $P(x) \geq$ grado $P_1(x)$

$Q(x)$ polinomio quoziente

$R(x)$ polinomio resto

$$\frac{x+1}{2x+3}$$

SOTTR.

$P(x)$ $\quad$ $P_1(x)$

$x+1$	$2x+3$
$x+\frac{3}{2}$	$\frac{1}{2}$
$= -\frac{1}{2}$	

$R(x)$ $\quad$ $Q(x)$

Allora

$$\frac{x+1}{2x+3} = \frac{1}{2} + \frac{-\frac{1}{2}}{2x+3} = \frac{1}{2} - \frac{1}{2}\,\frac{1}{2x+3}$$

Integriamo

$$\int \frac{x+1}{2x+3}\,dx = \int \frac{1}{2}\,dx - \frac{1}{2}\int \frac{1}{2x+3}\,dx$$

FATTO PRIMA

$$= \frac{1}{2}x - \frac{1}{2}\cdot\frac{1}{2}\ln|2x+3| + c$$

$$= \frac{1}{2}x - \frac{1}{4}\ln|2x+3| + c$$

· METODO DEI FRATTI SEMPLICI

$$\int \frac{1}{x^2-1}\,dx$$

Il denominatore è un polinomio di II grado con $\Delta > 0$ (2 SOL REALI)

$x^2-1 = (x-1)(x+1)$

Scriviamo $\frac{1}{x^2-1}$ in modo diverso

$$\int \frac{1}{(x^2-1)} = \frac{1}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1}$$

$$= \frac{A(x-1)+B(x+1)}{(x+1)(x-1)} = \frac{Ax - A + Bx + B}{(x+1)(x-1)} = \frac{x(A+B) - A + B}{x^2-1}$$

$$x\,(A+B) - A + B = 1 = 0\cdot x + 1$$

$$\begin{cases} A+B=0 \\ -A+B=1 \end{cases} \qquad \begin{cases} A=-B \\ 2B=1 \end{cases} \qquad \begin{cases} A=-\frac{1}{2} \\ B=\frac{1}{2} \end{cases}$$

Allora

$$\frac{1}{x^2-1} = -\frac{1}{2}\cdot\frac{1}{x+1} + \frac{1}{2}\,\frac{1}{x-1}$$

Integrando

$$\int \frac{1}{x^2-1}\,dx = -\frac{1}{2}\int \frac{1}{x+1}\,dx + \frac{1}{2}\int \frac{1}{x-1}\,dx$$

$$= -\frac{1}{2}\ln|x+1| + \frac{1}{2}\ln|x-1| + c$$

$$\cdot \int \frac{x+2}{x^2-5x+6}\,dx$$

IL METODO DEI FRATTI SEMPLICI SI APPLICA SE IL GRADO DEL NUMERATORE È MINORE DEL GRADO DEL DENOMINATORE

$$x^2 - 5x + 6 = 0$$

$$x_{1,2} = \frac{5 \pm \sqrt{25-24}}{2} = \frac{5 \pm 1}{2} = \begin{cases} 3 \\ 2 \end{cases}$$

Allora

$$(x^2 - 5x + 6) = (x-2)(x-3)$$

$$\frac{x+2}{x^2-5x+6} = \frac{A}{x-2} + \frac{B}{x-3} =$$

$$= \frac{A(x-3)+B(x-2)}{(x-2)(x-3)} = \frac{Ax-3A+Bx-2B}{x^2-5x+6} =$$

$$= \frac{x(A+B)-3A-2B}{x^2-5x+6}$$

Abbiamo ottenuto che

$$\frac{x+2}{x^2-5x+6} = \frac{x(A+B)-3A-2B}{x^2-5x+6}$$

Le due frazioni sono uguali a

$$x+2 = x(A+B) - 3A - 2B$$

$$\begin{cases} A+B=1 \\ -3A-2B=2 \end{cases} \qquad \begin{cases} A=1-B \\ -3(1-B)-2B=2 \end{cases}$$

$$\begin{cases} A=1-B \\ -3+3B-2B=2 \end{cases} \qquad \begin{cases} A=1-5=-4 \\ B=5 \end{cases}$$

Allora

$$\frac{x+2}{x^2-5x+6} = -\frac{4}{x-2} + \frac{5}{x-3}$$

Интеграл

$$\int \frac{x+2}{x^2-5x+6} = -4\int \frac{1}{x-2}\,dx + 5\int \frac{1}{x-3}\,dx$$

$$= -4\ln|x-2| + 5\ln|x-3| + C$$

$$\cdot \int \frac{x^2+1}{x^2-x}\,dx$$

ДЕЛЕНИЕ